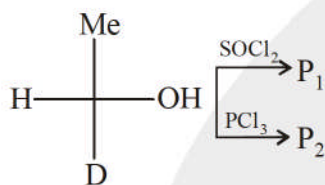


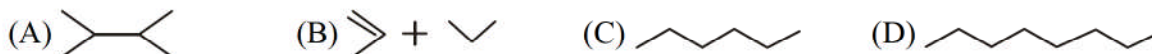
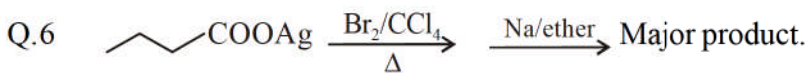
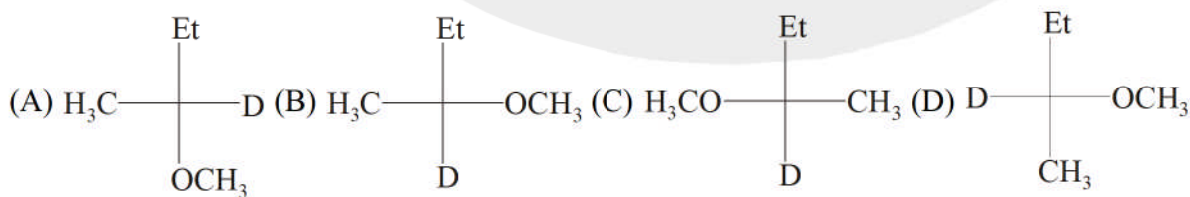
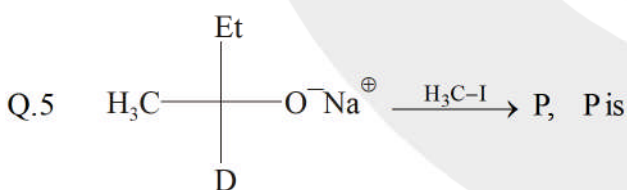
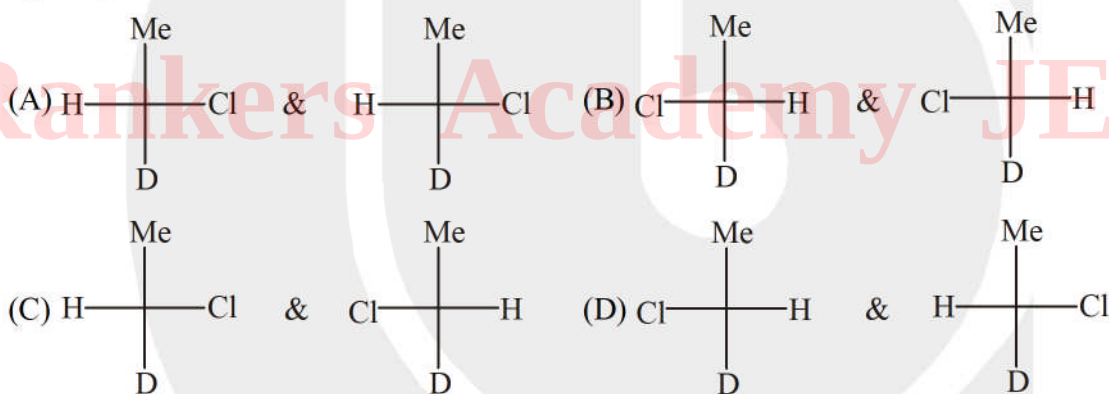
WORK SHEET - 06

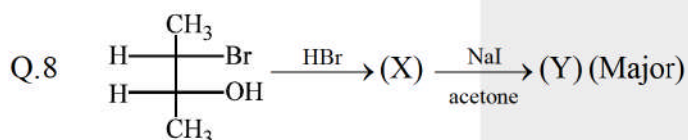
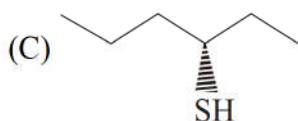
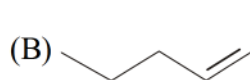
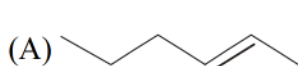
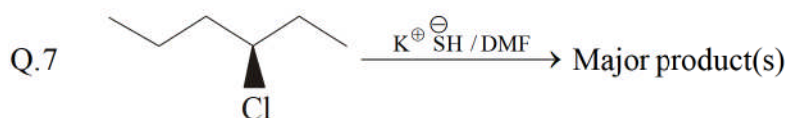
Single Correct

- Q.1 True statements regarding S_N1 reaction is/are
(A) Perfect racemisation is observed (B) Only Walden inversion is observed
(C) Total retention of configuration is observed (D) None of these
- Q.2 Pick an ether which cannot be prepared by direct Williamson's synthesis.
(A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OCH}_2\text{CH}_2\text{CH}_3$ (B) $\text{Ph-O-CH}_2\text{CH}_3$
(C) $(\text{CH}_3)_3\text{C-O-C}_2\text{H}_5$ (D) $\text{CH}_3\text{CH=CH-O-CH=CH}_2$
- Q.3 Which of the alkyl halides will under go S_N1 reaction at a fastest rate?
(A) $\text{Cl-CH}_2\text{-CN}$ (B) $\text{Cl-CH}_2\text{-NO}_2$ (C) $\text{Cl-CH}_2\text{-OMe}$ (D) $\text{Cl-CH}_2\text{-CH}_3$
- Q.4 Select the correct product from the following



P_1 & P_2 are respectively.





Product (Y) is

(A) cis-2-butene

(B) trans-2-butene

(C) 1-butene

(D) Iso-butene

Comprehension

Paragraph for question nos. 9 to 11

Wurtz reaction



Wurtz reaction take place via free radical mechanism as well as ionic mechanism. Generally major product will form by radical combination reaction if radical is 1° or 2° and major product will form by disproportionation reaction if radical is 3° . If diradical generate during reaction then generally major product form by intramolecular combination.

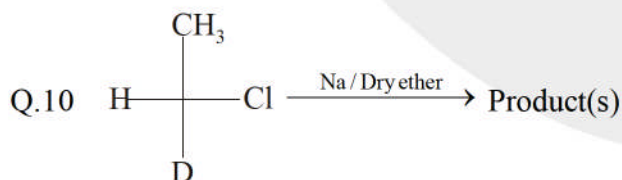
Q.9 Which statement is incorrect about Wurtz reaction.

(A) Reactivity order for halides is $\text{R-I} > \text{R-Br} > \text{R-Cl} > \text{R-F}$

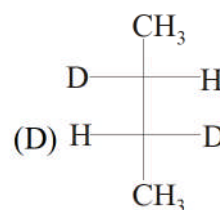
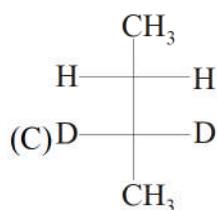
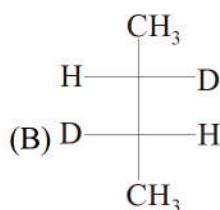
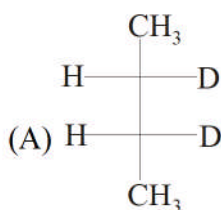
(B) If two type of halides are used then cross products also form during reaction.

(C) If $\text{H}_2\text{C}(\text{OH})-\text{CH}_2-\text{CH}_2-\text{Cl}$ is used during reaction then major product will be 

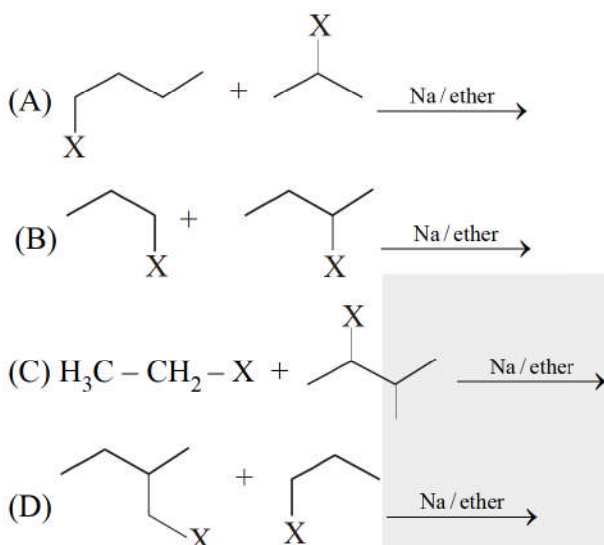
(D) Na metal acts as oxidising agent during reaction.



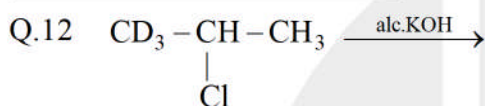
Which product should not form during reaction.



Q.11 3-Methylhexane can be prepared by



More than one may be correct



True about above reaction is/are:

- (A) It is E^2 elimination
 (B) Saytzeff's alkene is major product in above reaction
 (C) $\text{CD}_3-\text{CH}=\text{CH}_2$ is major product
 (D) It is anti elimination

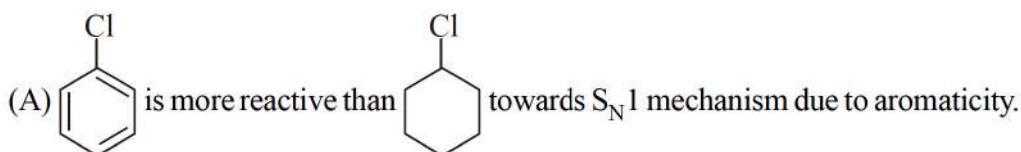
Q.13 Which of the following can give ppt. of AgCl with both aq. AgNO_3 & alc. AgNO_3 .



Q.14 Methane will be produced in which of the following reaction:



Q.15 Which of the statement is correct ?



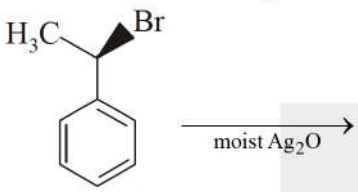
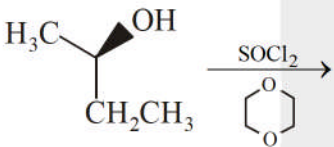
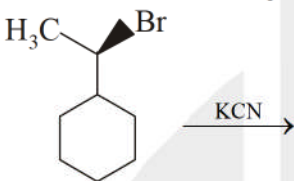
- (B) Inversion of configuration take place during $\text{S}_{\text{N}}2$ mechanism at α -chiral carbon.
 (C) Formation of $\text{R}-\text{Cl}$ from $\text{R}-\text{OH}$ by reaction with SOCl_2 is best method because byproducts are gases.
 (D) CH_4 can be prepared by decarboxylation of carboxylic acid.

Match the column

Q.16

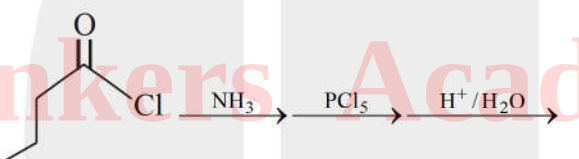
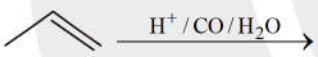
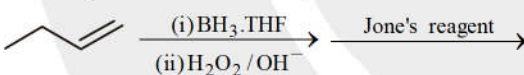
**Column I
(Reaction)**

**Column II
(Mechanism of reaction for major product)**

- | | |
|---|------------------------------------|
| (A) $\text{Me}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} + \text{MeMgCl} \longrightarrow$ | (P) $\text{S}_{\text{N}}1$ |
| (B)  | (Q) $\text{S}_{\text{N}}2$ |
| (C)  | (R) $\text{S}_{\text{N}}\text{i}$ |
| (D)  | (S) $\text{S}_{\text{N}}\text{AE}$ |
| | (T) $\text{S}_{\text{N}}\text{EA}$ |

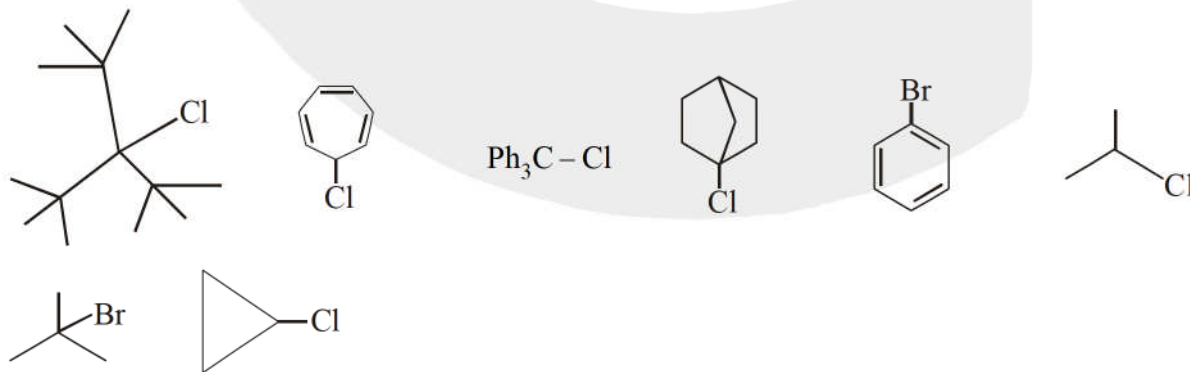
Subjective

Q.17 Consider following reactions:

- (a) 
- (b) 
- (c) 
- (d) 
- (e) 

Number of reactions which can produce Butyric acid as major product is / are ____.

Q.18 Consider following compounds.



Number of compounds in above compound which are more reactive than  Cl for $\text{S}_{\text{N}}1$ reaction is / are ____.

ANSWER KEY

Q.1	D	Q.2	D	Q.3	C	Q.4	C	Q.5	B
Q.6	C	Q.7	C	Q.8	B	Q.9	D	Q.10	C
Q.11	B	Q.12	ACD	Q.13	ABC	Q.14	ACD	Q.15	BCD
Q.16	[(A) S, (B) P, (C) R, (D) Q]			Q.17	[3]	Q.18	[4]		

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